إعداد الطالبة :- بتول هاتف جادر

مصفوفة ثنائية البعد

حل التمارين باستخدام القارئ Scanner

هذا من فضل ربي

JavaApplication8 - NetBeans IDE 8.2

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<default config>

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- Source Packages
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    - JavaApplication8.java
- Test Packages
- Libraries
- Test Libraries

Start Page | JavaApplication8.java

Source History

```java
package javaapplication8;
import java.util.Scanner;
public class JavaApplication8 {
public static void main(String[] args) {
 Scanner a=new Scanner(System.in);
    int x[][]=new int [3][3];
   System.out.print("Enter the number:");
    for(int i=0;i<3;i++)
    for(int j=0;j<3;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
}}}
```

Output - JavaApplication8 (run)

```
run:
Enter the number:
1 2 3
4 5 6
5 6 7
Print 2D arrayes :
1 2 3
4 5 6
5 6 7
BUILD SUCCESSFUL (total time: 7 seconds)
```

ادخال وطباعة عناصر مصفوفة 2D Array

11:22 INS

84°F Clear ENG 9:45 PM

JavaApplication8 - NetBeans IDE 8.2

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Search (Ctrl+I)

<default config>

Files Ser...

aApplication1
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aApplication3
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Start Page | JavaApplication8.java

Source History

```java
package javaapplication8;
import java.util.Scanner;
public class JavaApplication8 {
public static void main(String[] args) {
 Scanner a=new Scanner(System.in);
    int x[][]=new int [3][3];
    int sum=0;
   System.out.print("Enter the number:");
    for(int i=0;i<3;i++)
    for(int j=0;j<3;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
        sum=sum+x[i][j];
}
    System.out.println("sum is : "+ sum);
}}
```

Output - JavaApplication8 (run)

```
run:
Enter the number:
1 2 3
2 3 4
5 6 7
Print 2D arrayes :
1 2 3
2 3 4
5 6 7
sum is : 33
BUILD SUCCESSFUL (total time: 8 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد جمع المصفوفة

24:1 INS

84°F Clear ENG 10:08 PM

JavaApplication8 - NetBeans IDE 8.2

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Search (Ctrl+I)

<default config>

Files | Ser...

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- aApplication8
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- Test Libraries

Start Page | JavaApplication8.java

Source | History

```java
package javaapplication8;
import java.util.Scanner;
public class JavaApplication8 {
public static void main(String[] args) {
 Scanner a=new Scanner(System.in);
    int x[][]=new int [3][3];
    int sum=0;
   System.out.print("Enter the number:");
    for(int i=0;i<3;i++)
    for(int j=0;j<3;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
        sum=sum+x[i][j];
}
    System.out.println("sum is : "+ sum);
    System.out.println("sqrat is : "+ Math.sqrt(sum));
}}
```

Output - JavaApplication8 (run)

```
run:
Enter the number:
1 2 3
3 4 5
5 6 7
Print 2D arrayes :
1 2 3
3 4 5
5 6 7
sum is : 36
sqrat is : 6.0
BUILD SUCCESSFUL (total time: 7 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد جمع المصفوفة
- ✓ وايجاد الجذر التربيعي للمجموع

24:3 INS

84°F Clear ENG 10:11 PM

JavaApplication8 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Search (Ctrl+I)

<default config>

Files | Ser...

- aApplication1
- aApplication2
- aApplication3
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- Source Packages
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Start Page | JavaApplication8.java

Source | History

```java
package javaapplication8;
import java.util.Scanner;
public class JavaApplication8 {
public static void main(String[] args) {
 Scanner a=new Scanner(System.in);
    int x[][]=new int [3][3];
    int sum=0;
   System.out.print("Enter the number:");
    for(int i=0;i<3;i++)
    for(int j=0;j<3;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
        sum=sum+x[i][j];
}
    System.out.println("sum is : "+ sum);
    System.out.println("sqrat is : "+ Math.sqrt(sum));
     System.out.println("avarege is : "+ sum/9);
}}
```

Output - JavaApplication8 (run)

```
run:
Enter the number:2
1 2 3
4 5 6
7 8 9
Print 2D arrayes :
2 1 2
3 4 5
6 7 8
sum is : 38
sqrat is : 6.164414002968976
avarege is : 4
BUILD SUCCESSFUL (total time: 8 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد جمع المصفوفة
- ✓ وايجاد الجذر التربيعي للمجموع
- ✓ وايجاد المعدل لعناصر المصفوفة

24:33 INS

84°F Clear ENG 10:13 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
     int max=x[0][0];
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++){
          if(max<x[i][j])
              max=x[i][j];
}}
  System.out.println("max is : "+ max);
    }}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
2 3 4 4
5 6 6 7
1 2 3 4
8 9 3 4
Print 2D arrayes :
2 3 4 4
5 6 6 7
1 2 3 4
8 9 3 4
max is : 9
BUILD SUCCESSFUL (total time: 14 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد أكبر عنصر للمصفوفة

JavaApplication10 - NetBeans IDE 8.2

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- Libraries
- Test Libraries
- Application9
- Source Packages
  - javaapplication9
    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```
1   package javaapplication10;
2   import java.util.Scanner;
3   public class JavaApplication10 {
4   public static void main(String[] args) {
5       Scanner a=new Scanner(System.in);
6       int x[][]=new int [4][4];
7      System.out.print("Enter the number:");
8       for(int i=0;i<4;i++)
9       for(int j=0;j<4;j++)
10      x[i][j]=a.nextInt();
11     System.out.println("Print 2D arrayes :");
12      for(int i=0;i<4;i++){
13      for(int j=0;j<4;j++)
14      System.out.print(x[i][j]+" ");
        System.out.println();
16      }
17       int min=x[0][0];
18      for(int i=0;i<4;i++){
19      for(int j=0;j<4;j++){
            if(min>x[i][j])
21              min=x[i][j];
22  }}
23    System.out.println("min is : "+ min);
24      }}
25
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
1 2 4 5
2 3 4 4
5 6 7 8
9 8 7 5
Print 2D arrayes :
1 2 4 5
2 3 4 4
5 6 7 8
9 8 7 5
min is : 1
BUILD SUCCESSFUL (total time: 19 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد أصغر عنصر للمصفوفة

20:18 INS

83°F Clear ENG 10:32 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
     int max=x[0][0],min=x[0][0];
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++){
          if(max<x[i][j])
             max=x[i][j];
          if(min>x[i][j])
             min=x[i][j];
}}
  System.out.println("max is : "+ max);
  System.out.println("min is : "+ min);
 }}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
1 2 3 4
4 5 6 6
4 5 6 2
3 8 9 6
Print 2D arrayes :
1 2 3 4
4 5 6 6
4 5 6 2
3 8 9 6
max is : 9
min is : 1
BUILD SUCCESSFUL (total time: 14 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وايجاد أكبر و أصغر عنصر للمصفوفة

JavaApplication10 - NetBeans IDE 8.2

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- Application8
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- Test Libraries
- Application9
- Source Packages
  - javaapplication9
    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }

    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
          if(i==j)
     System.out.println(x[i][j]);
    else
  System.out.print(" ");
  System.out.println("");
 }}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
1 2 3 4
4 5 6 6
2 3 4 5
4 5 5 7
Print 2D arrayes :
1 2 3 4
4 5 6 6
2 3 4 5
4 5 5 7
1

 5

  4

   7

BUILD SUCCESSFUL (total time: 14 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وطباعة القطر الرئيسي فقط

26:1 INS

83°F Clear ENG 10:39 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Search (Ctrl+I)

<default config>

Files | Ser...

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- Application10
- Source Packages
  - javaapplication10
    - JavaApplication10.java
- Test Packages
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- Test Libraries
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- Application4
- Application5
- Application6
- Application7
- Application8
- Source Packages
  - javaapplication8
    - JavaApplication8.java
- Test Packages
- Libraries
- Test Libraries
- Application9
- Source Packages
  - javaapplication9
    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }

    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
          if(i+j==3)
     System.out.println(x[i][j]);
    else
  System.out.print(" ");
  System.out.println("");
 }}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
1 2 3 4
5 6 7 8
9 8 7 6
1 2 4 5
Print 2D arrayes :
1 2 3 4
5 6 7 8
9 8 7 6
1 2 4 5
   4

  7

 8

1

BUILD SUCCESSFUL (total time: 15 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وطباعة القطر الثانوي فقط

20:20 INS

83°F Clear ENG 10:41 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }

    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
        if(i==j)
      System.out.print(x[i][j]+" ");
          else if(i+j==3)
     System.out.print(x[i][j]);
    else
  System.out.print(" ");
  System.out.println("");
 }}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
1 2 3 4
5 6 7 7
4 5 6 7
3 4 5 6
Print 2D arrayes :
1 2 3 4
5 6 7 7
4 5 6 7
3 4 5 6
1    4
 6 7
 56
3   6
BUILD SUCCESSFUL (total time: 13 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ وطباعة القطر الرئيسي والثانوي

JavaApplication10 - NetBeans IDE 8.2

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Search (Ctrl+I)

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- Application8
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  - javaapplication8
    - JavaApplication8.java
- Test Packages
- Libraries
- Test Libraries
- Application9
- Source Packages
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    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++){
        if(i==j)
         x[i][j]=1;
    }}
    System.out.println("----------------");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
        System.out.println();
}}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
3 4 5 6
5 6 7 7
2 3 4 5
7 6 5 4
Print 2D arrayes :
3 4 5 6
5 6 7 7
2 3 4 5
7 6 5 4
----------------
1 4 5 6
5 1 7 7
2 3 1 5
7 6 5 1
BUILD SUCCESSFUL (total time: 13 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ استبدال القطر الرئيسي بال 1

16:6 INS

83°F Clear 10:50 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Search (Ctrl+I)

<default config>

Files Ser...

- Application1
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- Source Packages
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    - JavaApplication10.java
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- Libraries
- Test Libraries
- Application2
- Application3
- Application4
- Application5
- Application6
- Application7
- Application8
- Source Packages
  - javaapplication8
    - JavaApplication8.java
- Test Packages
- Libraries
- Test Libraries
- Application9
- Source Packages
  - javaapplication9
    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++){
        if(i+j==3)
         x[i][j]=1;
    }}
    System.out.println("----------------");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
        System.out.println();
}}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
2 3 4 5
7 6 5 4
2 4 5 6
9 6 4 3
Print 2D arrayes :
2 3 4 5
7 6 5 4
2 4 5 6
9 6 4 3
----------------
2 3 4 1
7 6 1 4
2 1 5 6
1 6 4 3
BUILD SUCCESSFUL (total time: 16 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ استبدال القطر الثانوي بال 1

19:18 INS

83°F Clear 10:50 PM

JavaApplication10 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Search (Ctrl+I)

<default config>

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- Test Libraries
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    - JavaApplication8.java
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- Test Libraries
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- Source Packages
  - javaapplication9
    - JavaApplication9.java
- Libraries

Start Page | JavaApplication10.java

Source | History

```java
package javaapplication10;
import java.util.Scanner;
public class JavaApplication10 {
public static void main(String[] args) {
    Scanner a=new Scanner(System.in);
    int x[][]=new int [4][4];
   System.out.print("Enter the number:");
    for(int i=0;i<4;i++)
    for(int j=0;j<4;j++)
    x[i][j]=a.nextInt();
   System.out.println("Print 2D arrayes :");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++){
        if(i+j<3)
         x[i][j]=1;
    }}
    System.out.println("----------------");
    for(int i=0;i<4;i++){
    for(int j=0;j<4;j++)
      System.out.print(x[i][j]+" ");
        System.out.println();
}}}
```

Output - JavaApplication10 (run)

```
run:
Enter the number:
2 3 4 5
5 6 7 8
3 4 5 6
9 7 5 4
Print 2D arrayes :
2 3 4 5
5 6 7 8
3 4 5 6
9 7 5 4
----------------
1 1 1 5
1 1 7 8
1 4 5 6
9 7 5 4
BUILD SUCCESSFUL (total time: 17 seconds)
```

- ✓ ادخال
- ✓ وطباعة عناصر مصفوفة
- ✓ استبدال المثلث العلوي بال 1

19:16 INS

83°F Clear ENG 10:53 PM